

Computer Organization And Assembly Language

```
00000000: 01001101 01011010 10010000 00000000 00000011 00000000 MZ....
00000006: 00000000 00000000 00000100 00000000 00000000 00000000 .....
0000000c: 11111111 11111111 00000000 00000000 10111000 00000000 .....
00000012: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000018: 01000000 00000000 00000000 00000000 00000000 00000000 @.....
0000001e: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000024: 00000000 00000000 00000000 00000000 00000000 00000000 .....
0000002a: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000030: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000036: 00000000 00000000 00000000 00000000 00000000 00000000 .....
0000003c: 10000000 00000000 00000000 00000000 00001110 00011111 .....
00000042: 10111010 00001110 00000000 10110100 00001001 11001101 .....
00000048: 00100001 10111000 00000001 01001100 11001101 00100001 !..L.!
0000004e: 01010100 01101000 01101001 01110011 00100000 01110000 This p
00000054: 01110010 01101111 01100111 01110010 01100001 01101101 rogram
0000005a: 00100000 01100011 01100001 01101110 01101110 01101111 canno
00000060: 01110100 00100000 01100010 01100101 00100000 01110010 t be r
00000066: 01110101 01101110 00100000 01101001 01101110 00100000 un in
0000006c: 01000100 01001111 01010011 00100000 01101101 01101111 DOS mo
00000072: 01100100 01100101 00101110 00001101 00001101 00001010 de....
00000078: 00100100 00000000 00000000 00000000 00000000 00000000 $. ....
0000007e: 00000000 00000000 01010000 01000101 00000000 00000000 ..PE..
```

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Objectives:

1. Understanding and working with string instructions

Lab Instructor:

Mr. Tariq Mehmood Butt tariq.butt@pucit.edu.pk

Teacher Assistants:

Hafiz Muhammad Ahmad

bcsf21m502@pucit.edu.pk

Syed Muhammad Zain Raza

bcsf21m510@pucit.edu.pk

Zahra Malik

bcsf21m551@pucit.edu.pk

Lab Tasks

Task 1: Write an Assembly Language Program that will copy a string STRING_1 into another string STRING_2 using the instruction MOVSB and REP, and also display it.

Task 2: Write an Assembly Language Program that prompts the user to enter a string, then stores it in a variable using instruction STOSB.

Task 3: Write two procedures for input and output, which are capable of getting and displaying data in any format(i.e. Binary, Octal, decimal or hexadecimal). The procedures should accept the output format as parameter and work accordingly.

→ Implement main procedure which:

- 1) Takes input in binary and displays output in octal, hexadecimal, and decimal formats.
- 2) Takes input in octal and displays output in binary, hexadecimal, and decimal formats.
- 3) Takes input in decimal and displays output in octal, hexadecimal, and binary formats.
- 4) Takes input in hexadecimal and displays output in binary, octal, and decimal formats.

Note: You are **not allowed** to use **mov, and, or, sub, shift, rotate** instructions in input and output procedures and also not allowed to use add as you used to do in previous.

Hint: Declare arrays of numbers and characters and use string instructions for manipulating data.